Nuclear Transmutation:

The conversion of one element into another is known as transmutation. During which one type of atom converts into another type. It may be natural as well as artificial. In natural transmutation certain radioactive elements present decay spontaneously by emitting alpha or beta particles. In artificial transmutation, bombardment of atoms take place with the help of high energy α-particle. The element Uranium goes spontaneous transmutation as

$$_{92}U^{238} \rightarrow _{90}Th^{234} + _{2}He^{4}$$

In above nuclear reaction/transmutation, Uranium nucleus disintegrates into Thorium nucleus with emission of Helium nucleus.

On the other hand, in artificial transmutation in Nitrogen isotope,

$$_{7}N^{14} + _{2}He^{4} \rightarrow _{8}O^{17} + _{1}H^{1}$$

$$_{4}Be^{9} + _{2}He^{4} \rightarrow _{6}C^{12} + _{0}n^{1}.$$

In above examples, positive ion has been employed as a bombarding agent for nuclear transmutation. In some cases neutron ~~is~~ also is used to bombard on the nucleus. It is most commonly used in the transmutation to form synthetic isotopes because neutrons are neutral, they are not repelled by nucleus.

e.g.

$$_{26}Fe^{58} + {}_{0}n^{1} \rightarrow {}_{26}Fe^{59} \rightarrow {}_{27}Co^{59} + {}_{-1}e^{0}.$$

The transmutation is also called disintegration because the transmuted element disintegrates forming a different and new element. The earliest experiments on artificial nuclear disintegration were performed by Ernest Rutherford.

Radioactivity:

The radioactivity was first discovered by Henri Becquerel in 1896. He found that radiations were given off spontaneously and not influenced by any external agencies. The term radioactivity involves the spontaneous transmutation of one element into another either naturally or artificially.

The phenomenon of spontaneous emission of highly penetrating radiations from heavy elements of atomic weights greater than about 206, occuring in nature is called natural radioactivity. Such elements are called radioactive elements. Radioactivity is uneffected by any external agents such as high temperature, high pressure, large electric or magnetic field. The nuclear radiation is composed of three distinct kinds as α & β particles & γ-radiation

Moreover, the modern technique of artificial transmutation of elements has been able to produce ~~in an element~~ radioactivity. The radioactivity produced in an element by bombarding it with α-particle, neutrons, protons & other particles or radiation is called artificial radioactivity.

When the nucleus of an atom emits an alpha particle, the atom is transformed into a new atom. It's atomic mass is decreased by four units & its atomic number is decreased by two units. For an example, Radon nucleus with atomic mass (A) = 226 and atomic number (Z) = 88 undergoes radioactive disintegration by emitting an α-particle. The new product will be having its nucleus with A = 222 & Z = 86.

$$_{88}Ra^{226} \rightarrow {}_{86}Rn^{222} + ({}_{2}He^{4}) \rightarrow {}_{84}Po^{218} + ({}_{2}He^{4}) + \cdots + {}_{82}Pb^{206}$$

(Radium) (Radon) (α) (Polonium) (α) (lead)

When there is emission of α-particle, change in nuclear structure takes place as follows.

$$_{Z}X^{A} \longrightarrow {}_{Z-2}X^{A-4} + \alpha\,(2p, 2n)$$

When a β- particle is emitted by a nucleus of atomic number Z, the atomic number of the new atom formed becomes Z+1 but mass of nucleus remains practically unaltered. Since mass of a β-particle is negligible in comparision to the mass of the nucleus. The parent and the product atoms form a pair of isobar

$$ {}_{Z}X^{A} \rightarrow {}_{Z+1}X^{A} + \beta $$

e.g. $ {}_{6}C^{14} \rightarrow {}_{7}N^{14} + {}_{-1}\beta^{0} $.

When beta decay occurs the nucleus emits the particle and changes into proton. $[n \rightarrow p^{+} + e^{-}]$

Gamma decay, it is a type of radioactivity in which some unstable atomic nuclei dissipate excess energy by a spontaneous electromagnetic process. During gamma emission energy of ~~extremely~~ extremely high value emitted constituting radiation of extremely short wavelength. The large value of energy associated with γ-rays show that they must be of nuclear origin. An excited nucleus can return to the ground state is by the emission of γ-rays.

$$ [{}_{Z}X^{A}]^{*} \rightarrow {}_{Z}X^{A} + \gamma $$

($* \rightarrow$ excited state)

The emission of gamma rays doesn't alter the number of protons or neutrons in the nucleus but instead has the effect of moving nucleus from a higher to a lower energy state. Paul Villard, a French chemist & physicist discovered gamma radiation in 1900 AD, while studying radiation emitted from Radium. Gamma rays have energy $h\nu = E^{*} - E$ where $\{E^{*}$ = excited state, E = normal state.

where ν = frequency of γ-rays.

Gamma rays are used in medicine (radiotherapy), industry (sterilization & disinfection) & the nuclear industry.

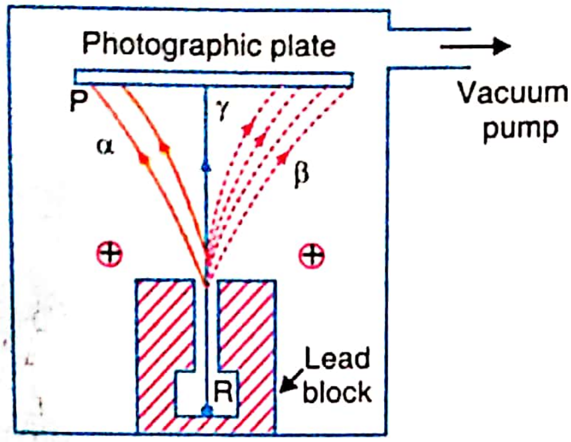


Fig. 31.1

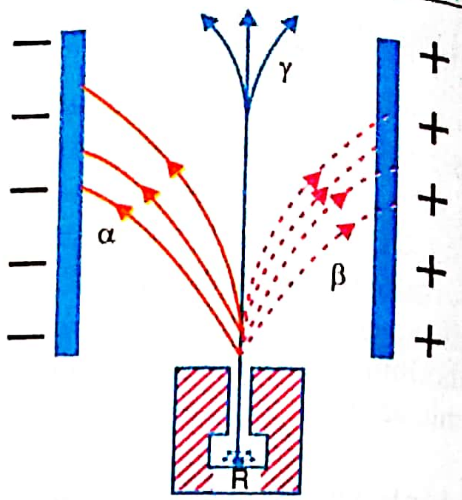


Fig. 31.2

31.4 Properties of Alpha Rays

(1) An α-particle is a helium nucleus consisting of two protons and two neutrons. It carries two units of positive charge.

(2) α-particles shot out from radioactive substances have high velocities, ranging from 1.4×10^7 to 1.7×10^7 m/s. They move along straight lines. Their tracks can be observed in the Wilson's cloud chamber.

(3) They produce intense ionisation in the gas through which they pass. Their ionising power is 100 times greater than that of β-rays and 10,000 times greater than that of γ-rays.

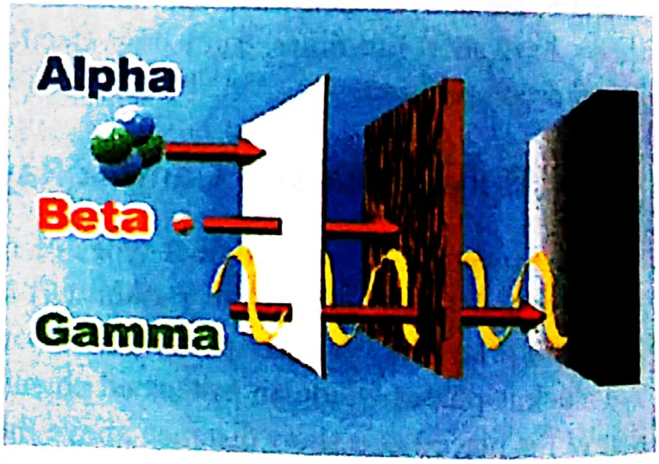
(4) They affect a photographic plate but the effect is very feeble.

(5) They produce fluorescence when they fall on substances like barium platinocyanide or zinc sulphide. The fluorescence can be observed through a low power microscope.

(6) They are deflected by electric and magnetic fields. This shows that they are charged particles.

(7) They are scattered by nuclei of heavy elements like gold.

(8) They produce a heating effect. The evolution of heat is due to the stoppage of α, β and γ-rays by the radioactive substance.



Properties of α, β, γ-Rays.

(9) The value of E/M for α-particles was found by Rutherford to be half of that for hydrogen ion. The charge on each α-particle (E) was found to be twice that of a hydrogen ion. Thus the mass of the α-particles was shown to be four times that of hydrogen. Therefore, an α-particle is the nucleus of the helium atom.

31.5 Properties of Beta Rays

(1) β-particles possess negative charge and mass equal to that of an electron.

(2) All the β-particles emitted from a substance do not have the same velocity. The velocities of the β-particles emitted by a substance range continuously from $0.3c$ to $0.99c$. At high velocities, e/m is found to decrease indicating an increase in mass of the particles, according to Einstein's equation,

$$m = \frac{m_0}{\sqrt{1 - v^2 / c^2}}.$$

(3) Their ionising power is low and hence, range is large.

(4) They affect a photographic plate.

(5) They produce fluorescence in barium platinocyanide, willemite etc.

(6) They are deflected by electric and magnetic fields. Their direction of deflection indicates that they are negatively charged particles.

(7) They penetrate through thin metal foils and their penetrating power is greater than that of α-rays.

(8) The value of e/m for β-rays was proved to be the same as that of cathode rays. The charge on a β-particle was observed to be the same as the charge on an electron. Therefore, β-rays are identical with electrons.

31.6 Properties of Gamma Rays

(1) γ-rays are electromagnetic waves of very short wavelength. The wavelengths of γ-rays range from 0.005 Å to 0.5 Å. They are not charged particles. They travel with the velocity of light.

(2) They produce fluorescence and affect a photographic plate.

(3) They ionise the gas through which they pass but the ionisation produced is very small.

(4) They are more penetrating than even β-rays. They can even pass through an iron plate of 30 cm thickness.

(5) They are not affected by electric and magnetic fields.

(6) They are diffracted by crystals like X-rays.

Law of Radioactive Disintegration:

Let N be the number of atoms present in a particular radioelement at a given instant 't'. Then the rate of decrease of the atom is $-\frac{dN}{dt}$. It is proportional to N.

$$\therefore -\frac{dN}{dt} \propto N \quad \text{or} \quad \frac{dN}{dt} = -\lambda N$$

λ = proportionality const
= decay const
= disintegration const.

$$\text{or} \quad \frac{dN}{N} = -\lambda dt \quad \text{---} (1)$$

on integrating eqn (1) $\int \frac{dN}{N} = -\lambda \int dt$

$$\text{or,} \quad \log_e N = -\lambda t + C \quad \text{---} (2)$$

where C is integration constant.

Suppose the number of atoms initially present in the radioelement be N_o. Then, when $t=0$, $N = N_o$.

So from eqn. (2) $C = \log_e N_o$

Now substituting value of C in eqn (2),

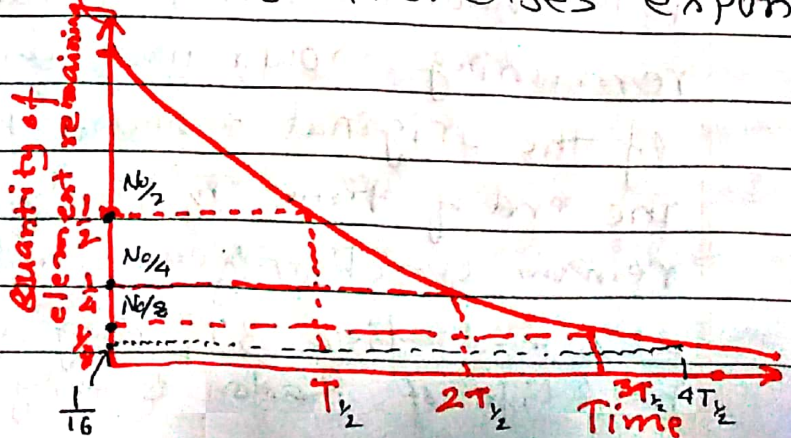
$$\log_e N = -\lambda t + \log_e N_o$$

$$\text{or} \quad \log\left(\frac{N}{N_o}\right) = -\lambda t$$

$$\frac{N}{N_o} = e^{-\lambda t} \quad \text{or} \quad N = N_o e^{-\lambda t} \quad \text{---} (3)$$

The relation (3) shows that the number of atoms of a given radioactive substance decreases exponentially with time.

Disintegration of element is $\frac{N_o}{2}$ in time T, $\frac{N_o}{4}$ in time $2T$, $\frac{N_o}{8}$ in time $3T$ .. and so on.
T is half life period.



Half Life Period (T):

The half life period of a radioactive element may be difined as the time during which a given amount of that element is reduced ~~by~~ by distintegration to half its initial amount. It is generally denoted as t or $T_{1/2}$

we have

$$N = N_0 e^{-\lambda t}$$

For $t = T_{1/2}$, $N = \frac{N_0}{2}$.

Then $\frac{N_0}{2} = N_0 e^{-\lambda T_{1/2}}$

or $e^{+\lambda T_{1/2}} = 2$

or $\lambda T_{1/2} = \log_e 2$ (Taking log bothsides)

or $T_{1/2} = \frac{\log_e 2}{\lambda} = \frac{0.6931}{\lambda}$.

So, half life Period $\boxed{T_{1/2} = \frac{0.6931}{\lambda}}$. --- (4)

As seen in the above relation, $T_{1/2}$ is inversely proportional to the radioactive constant and hence is a characteristic constant which can be employed to differentiate radioelements.

For an example, it takes nearly 1620 years for one half of a given quantity of radium (Ra) to change into radon (Rn). In another 1620 years half of the remaining would have disintegrated leaving one-forth of the original amount behind. In other words at the end of time $T_{1/2}$, only 50% of the radiactive atoms remain unchanged. Similarly, at the end of $2T_{1/2}$ only 25% and after $3T_{1/2}$ only 12.5% is remaing behind. The half life of Radon is only 3.8 days.

Radioactive Decay Constant (λ):

From the eqn ①, above,

$$\lambda = -\frac{dN/dt}{N}$$

from the relation, decay constant may be defined as the ratio of the amount of the substance which disintegrates in a unit time to the amount of the substance present.

Suppose, we put $t = 1/\lambda$ in eqn. ③ above, then the number of the original atoms present after this time is $N = N_o e^{-\lambda t} = N_o e^{-\lambda \cdot \frac{1}{\lambda}}$

$$\text{or, } N = N_o e^{-1}$$

$$\text{or, } N = \frac{N_o}{e} = \frac{N_o}{2.718}$$

$$\text{or } N = N_o\, 0.368$$

$$\text{or } N = N_o (0.37) \quad \text{--- (5)}$$

$$\text{or } N = (37\%) N_o.$$

Hence radioactive constant may also be defined as the reciprocal of the time during which the number of atoms of a radioactive substance falls to 37% of its original value.

The Mean Life (Average Life): ($\bar{T}$)

It is not possible to predict which atom of a radioactive substance will disintegrate at any instant. The atom which disintegrates first has zero life and that disintegrat last has infinite life. Thus the life of every atom is different and the actual lives of the various atoms range from zero to infinity.

$$\therefore \text{ Mean life} = \frac{\text{Sum of lives of all the atoms}}{\text{Total number of atoms.}} \quad -(6)$$

Definition: The mean life of a radioelement is defined as the ratio of the total life time of all the radioactive atoms to the total number of such atoms in it.

Value of Mean life:

Let N_0 be the total number of radioactive atoms in the beginning. Let N be the number of atoms of that element after time t. Then $N = N_0 e^{-\lambda t}$.
Let dN be the number of atoms disintegrating between time t and $t+\delta t$. These dN atoms have had a life between t and $(t+\delta t)$, where δt is very small time. Neglecting δt, each of these atoms had a life of t.
Then total life of dN atoms $= (dN)\, t$.

The possible life of any of the total number N_0 radioactive atoms varies from 0 to ∞.
Total life time of all N_0 atoms $= \int_0^\infty t\, dN$.
Let mean ~~life~~ life be denoted as $\bar{T}$
Then $\bar{T} = \frac{\text{total life time}}{\text{total number of atoms}}$

or $$\bar{T} = \frac{\int_0^\infty t\, dN}{N_0}$$

We have $N = N_0 e^{-\lambda t}$

$$\frac{dN}{dt} = \frac{d}{dt}\left(N_0 e^{-\lambda t}\right)$$

$$= N_0 \cdot (-\lambda)\, e^{-\lambda t}$$

$$dN = -\lambda N_0 e^{-\lambda t}\, dt$$

Leaving the negative sign which simply indicates the decrease in the number of atoms with time

So, $dN = \lambda N_0 e^{-\lambda t}$.

$$\therefore \quad \bar{T} = \frac{\int_0^\infty t \cdot \lambda N_0 e^{-\lambda t}}{N_0} = \lambda \int_0^\infty t\, e^{-\lambda t}\, dt.$$

Now integrating by parts,

$$\bar{T} = \lambda \left[t \cdot \frac{e^{-\lambda t}}{-\lambda} - \int \frac{e^{-\lambda t}\, dt}{-\lambda} \right]_0^\infty$$

$$= \lambda \left[\frac{t e^{-\lambda t}}{-\lambda} - \frac{e^{-\lambda t}}{-\lambda^2} \right]_0^\infty$$

$$= \lambda \cdot \frac{1}{\lambda^2} \left[-t e^{-\lambda t} \lambda - e^{-\lambda t} \right]_0^\infty$$

$$= -\frac{1}{\lambda} \left[(t\lambda + 1)\, e^{-\lambda t} \right]_0^\infty$$

$$= -\frac{1}{\lambda} \left[(t\lambda + 1)\, e^{-\lambda t} \right]_0^\infty = -\frac{1}{\lambda}(-1) \quad (*)$$

(*) $\bar{T} = -\frac{1}{\lambda}\left[(t\lambda+1)e^{-\lambda t}\right]_0^\infty$

$= -\frac{1}{\lambda} \lim_{k\to\infty} \left[(t\lambda+1)e^{-\lambda t}\right]_0^k$

$= -\frac{1}{\lambda} \lim_{k\to 0} \left[(k\lambda+1)e^{-k\lambda} - (0\cdot\lambda+1)e^{-\lambda \cdot 0}\right]$

$= -\frac{1}{\lambda} \left[\lim_{k\to 0} (k\lambda+1)e^{-k\lambda} - \lim_{k\to\infty} e^{0}\right]$

$= -\frac{1}{\lambda}(0-1) = \frac{1}{\lambda}$

$$\bar{T} = \frac{1}{\lambda} \quad \text{------} \ (7),$$

Obviously, from above relation (7), average life or life expectancy of an individual radioactive atom is equal to the reciprocal of the decay constant.

Also, $T_{1/2} = \frac{0.693}{\lambda} = 0.693\,\bar{T} \Rightarrow \boxed{T_{1/2} = 0.693\,\bar{T}}$ — (8)

Thus half life period is equal to 69.3% of average life.

Law of Successive Disintegration:—

The chain disintegration means a process in which a radioactive substance disintegrates to form a new substance, which disintegrates to form another new substance and so on. Consider a substance A which decays to form a new substance B. The substance B decays to C and so on.

i.e. $A \rightarrow B \rightarrow C \rightarrow \cdots\cdots X$ (stable)

In a radioactive series, any two adjacent elements are considered as parent & daughter evidently in a series, the parent of the following element will be daugher of the preceding one.

At $t=0$, the number of initial atoms in $A = N_0$

the number of initial atoms in $B = 0$

At time $t=t$, let number of atoms in $A = N_1$

number of atoms in $B = N_2$

Let λ_1 and λ_2 be the decay constants of A & B respectively. Every time an atom of A disappears, then an atom of B is produced.

$\therefore$ Rate of formation of B from $A = \lambda_1 N_1$

The rate at which B disintegrates $= \lambda_2 N_2$.

The net increase in the number of B atoms

$$\frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2$$

Hence $\frac{dN_2}{dt} = \lambda_1 N_0 e^{-\lambda_1 t} - \lambda_2 N_2 \quad (\because N_1 = N_0 e^{-\lambda_1 t})$

$$\therefore \frac{dN_2}{dt} + \lambda_2 N_2 = \lambda_1 N_0 e^{-\lambda_1 t}$$

Multiplying both sides by the integrating factor $e^{\lambda_2 t}$,

$$\frac{dN_2}{dt}\cdot e^{\lambda_2 t} + \lambda_2 N_2 e^{\lambda_2 t} = \lambda_1 N_0 e^{(\lambda_2-\lambda_1)t}$$

or, $$\frac{d}{dt}\left[N_2 e^{\lambda_2 t}\right] = \lambda_1 N_0 e^{(\lambda_2-\lambda_1)t}$$

Integrating, $$N_2 e^{\lambda_2 t} = \frac{\lambda_1}{\lambda_2-\lambda_1}\cdot N_0 e^{(\lambda_2-\lambda_1)t} + C \quad \text{--(1)}$$

C = integrating constant.

The constant of integration C can be determined from the initial condition, as,

when $t=0$, $N_2=0$ $\therefore$ $$C = -\frac{\lambda_1 N_0}{\lambda_2-\lambda_1} \quad \text{--(2)}$$

Now substituting value of c in eqn (1) from eqn (2),

$$N_2 e^{\lambda_2 t} = \frac{\lambda_1 N_0}{\lambda_2-\lambda_1}\left(e^{(\lambda_2-\lambda_1)t} - 1\right)$$

or $$N_2 = \frac{\lambda_1 N_0}{\lambda_2-\lambda_1}\left[e^{-\lambda_1 t} - e^{\lambda_2 t}\right].$$

Soddy Fajan's Displacement law:

When a radioactive disintegration occurs with the emission of α and β particles, the original atom is called the parent atom changes into something else, called the daughter. In 1913 Soddy and Fajans discovered a simple law known as the displacement law which governs these radioactive changes.

They can be stated as follows–

1) In all known radioactive transformations either an α- or a β- particle (i.e. never both or more than one of each kind) is emitted by the atom.

2) When a radioactive atom emits an α-particle, a new atom is formed whose mass number is less by four units & atomic number less by two units than those of the parent atom.

e.g. $_{92}U^{238} \xrightarrow{-\alpha} {}_{90}Th^{234}$

3). When a radioactive atom emits a β-particle, the new atom formed has the same mass number but the atomic number is increased by one unit.

e.g. $_{90}Th^{234} \xrightarrow{-\beta} {}_{91}Pa^{234}$.

Natural Radioactive Series:-

With the help of the displacement law, one can determine easily the mass and atomic numbers of the different elements in the successive radioactive change, if the mass and atomic numbers of the parent are known. The different atoms produced at the different stages of disintegration will naturally form a series.

The first member of the series is called the Parent, the intermediate member is called the daughter and the final stable member is called the end product. There are four radioactive series.

1) Uranium Series. $[{}_{92}U^{238}]$
2) Actinium series $[{}_{92}U^{235}]$
3) Thorium Series $[{}_{90}Th^{232}]$
4) Neptunium series. $[{}_{93}Np^{237}]$.

Each of the series consists of a succession of daughter elements which are all ~~derived~~ derived from a single parent nuclide. We know that alpha decay reduces the mass number of a nucleus by 4 unit. Thus the nuclides whose mass numbers are all given by $A = 4n$, where n is an integer, can decay into one another in descending order of mass number. All radioactive nuclides whose mass numbers obeys the equation $A = 4n$ are said to be members of the $4n$ series. Similarly, radioactive nuclides whose mass numbers obey the equation $A = 4n+1$ belong to the $(4n+1)$ series. The mass number satisfying the equations $A = (4n+2)$ and $A = (4n+3)$ belonging to the $(4n+2)$ series and $(4n+3)$ series respectively. Following table represents a list of the names of four important radioactive series, their parent includes and the half life/lives of these parents, and the stable daughters which are end products of the series.

Four Radioactive Series

S.N	Mass number	Series	Parent	Half life (Years)	end product.
1	$4n$	Thorium	${}_{90}Th^{232}$	1.39×10^{10}	${}_{82}Pb^{206}$
2	$4n+1$	Neptunium	${}_{93}Np^{237}$	2.25×10^{6}	${}_{83}Bi^{209}$
3	$4n+2$	Uranium	${}_{92}U^{238}$	4.51×10^{9}	${}_{82}Pb^{206}$
4	$4n+3$	Actinium	${}_{92}U^{235}$	7.07×10^{8}	${}_{82}Pb^{207}$

Let us consider one of the series, say the Uranium series. In this case the parent element is Uranium with mass number $A = 238$ & $Z = 92$. It is also known as Uranium I or UI. It ($_{92}UI^{238}$) emits an α-particle and the product formed is Uranium X_1 with $Z = 92 - 2 = 90$ and $A = 238 - 4 = 234$. UX_1 emits a β-particle and the product formed is UX_2 with $Z = 90 + 1 = 91$ and $A = 234$. The nucleus UX_2 emits a β-particle and the product formed is UII with $Z = 92$ $(= 91 + 1)$ and $A = 234$. Thus UII is an isotope of UI. The final stable ~~product~~ end product is lead $_{82}Pb^{206}$ ($_{82}Pb^{206}$).

Table 11.2. Uranium Series

No.	Substance	Symbol	Atomic Number Z	Mass Number A	Particles Emitted	Half-Life Period T
1	Uranium I	UI	92	238	α	$4{\cdot}56 \times 10^9$ years
2	Uranium X_1	UX_1	90	234	β, γ	24·1 days
3	Uranium X_2	UX_2	91	234	β, γ	1·18 minutes
4	Uranium II	UII	92	234	α	$2{\cdot}56 \times 10^5$ years
5	Ionium	Io	90	230	α, γ	$8{\cdot}3 \times 10^4$ year
6	Radium	Ra	88	226	α, γ	1622 years
7	Radon	Rn	86	222	α	3·825 days
8	Radium A	RaA	84	218	α, γ	3·05 minutes
9	Radium B	RaB	82	214	β, γ	26·8 minutes
10	Radium C	RaC	83	214	α or β, γ	19·7 minutes
11a	Radium C′	RaC′	84	214	α	1.6×10^{-4} second
	or	or				
11b	Radium C″	RaC″	81	210	β	1·32 minutes
12	Radium D	RaD	82	210	β, γ	22 years
13	Radium E	RaE	83	210	β	5 days
14	Radium F (Polonium)	RaF	84	210	α, γ	138·3 days
15	Radium G (Lead)	Pb (RaG)	82	206	Stable	**Lead A**=206

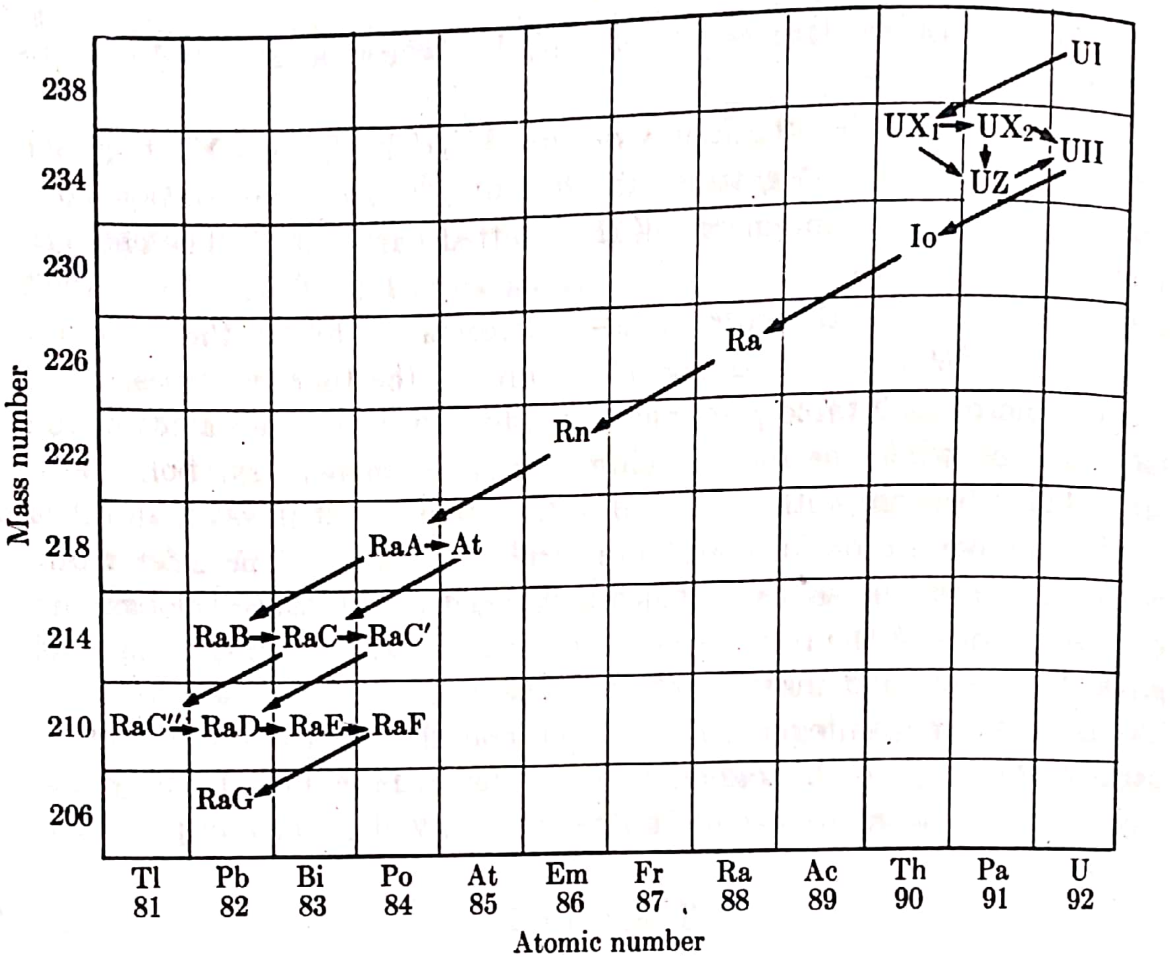


FIG. 10–10. The uranium (4n + 2) series.

TABLE 10–3

THE THORIUM SERIES

Radioactive species	Nuclide	Type of disinte-gration	Half-life	Disinte-gration constant, sec^{-1}	Particle energy, Mev
Thorium (Th)	$_{90}Th^{232}$	α	1.39×10^{10} y	1.58×10^{-18}	4.007
Mesothorium1 (MsTh1)	$_{88}Ra^{228}$	β	6.7 y	3.28×10^{-9}	0.04
Mesothorium2 (MsTh2)	$_{89}Ac^{228}$	β	6.13 h	3.14×10^{-5}	2.18
Radiothorium (RdTh)	$_{90}Th^{228}$	α	1.910 y	1.15×10^{-8}	5.423 m
Thorium X (ThX)	$_{88}Ra^{224}$	α	3.64 d	2.20×10^{-6}	5.681 m
Th Emanation (Tn)	$_{86}Em^{220}$	α	51.5 s	1.34×10^{-2}	6.280
Thorium A (ThA)	$_{84}Po^{216}$	α, β	0.16 s	4.33	6.774
Thorium B (ThB)	$_{82}Pb^{212}$	β	10.6 h	1.82×10^{-5}	0.58
Astatine-216 (At^{216})	$_{85}At^{216}$	α	3×10^{-4} s	2.3×10^{3}	7.79
Thorium C (ThC)	$_{83}Bi^{212}$	α, β	60.5 m	1.91×10^{-4}	α:6.086 m β:2.25
Thorium C′ (ThC′)	$_{84}Po^{212}$	α	3.0×10^{-7} s	2.31×10^{6}	8.780
Thorium C″ (ThC″)	$_{81}Tl^{208}$	β	3.10 m	3.73×10^{-3}	1.79
Thorium D (ThD)	$_{82}Pb^{208}$	Stable			

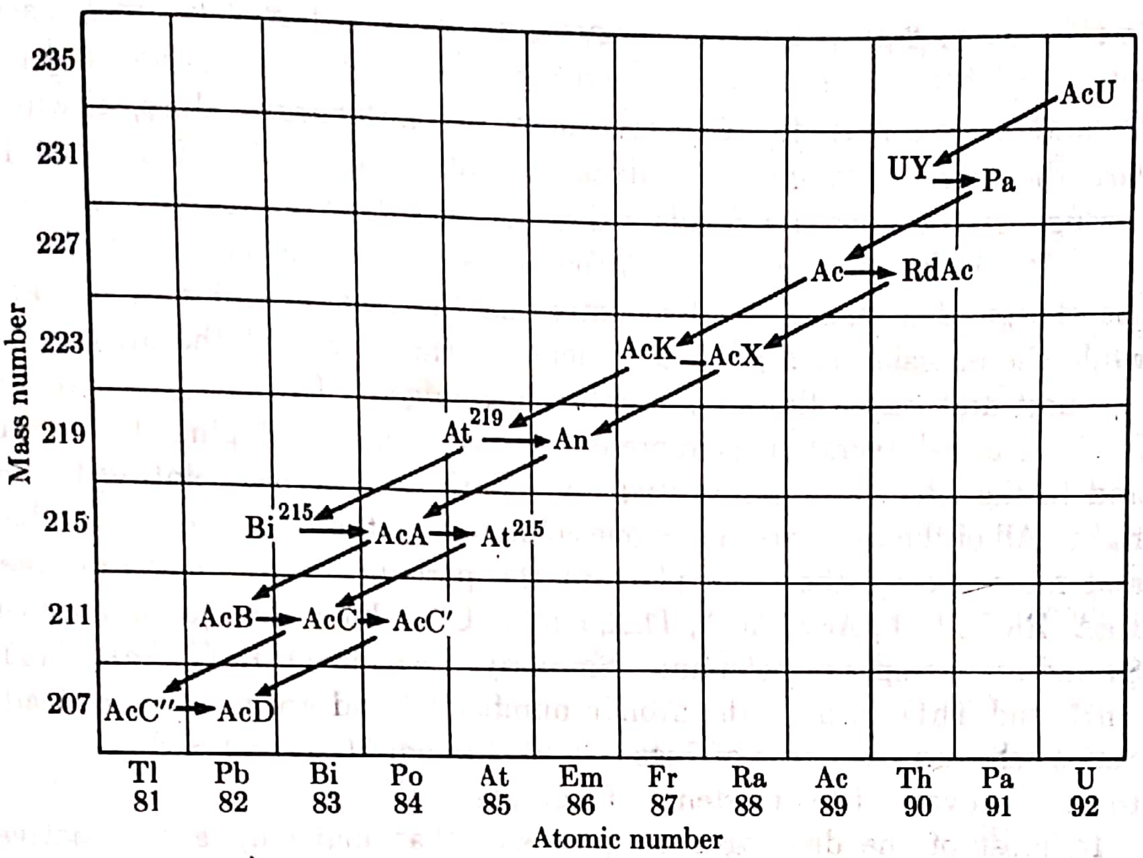


FIG. 10–11. The actinium (4n + 3) series.

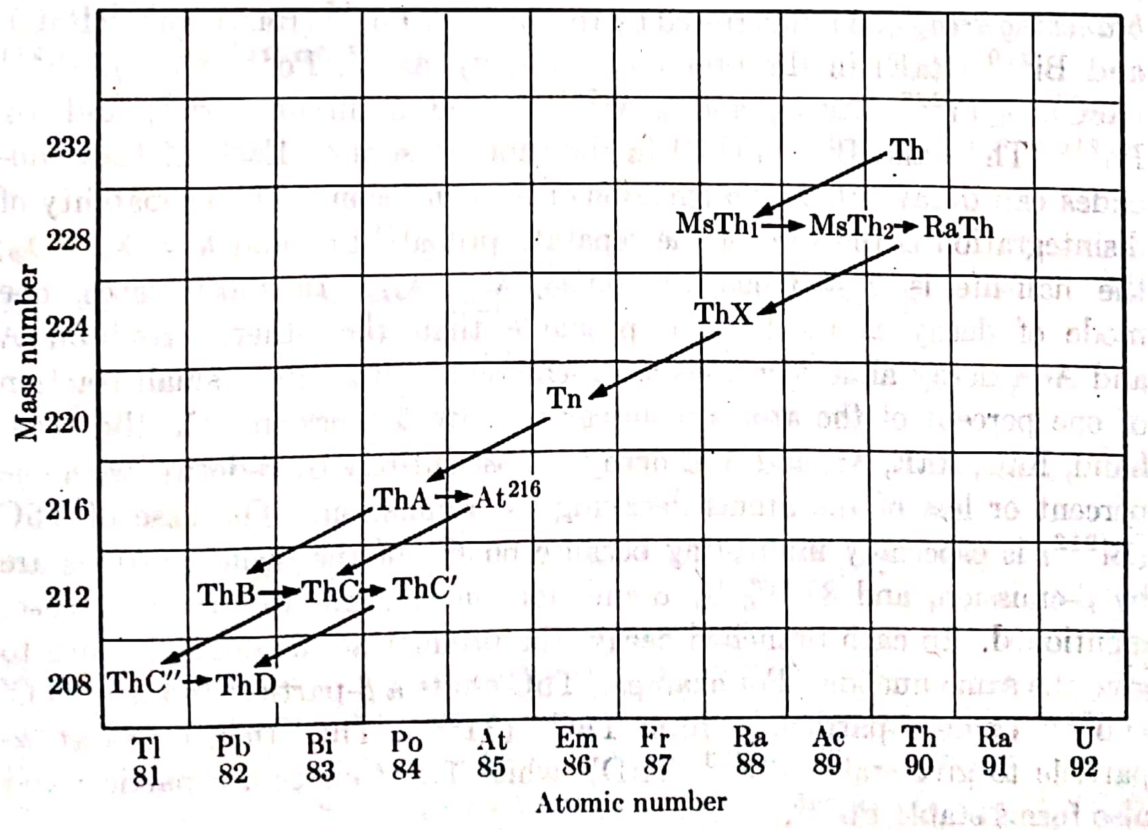


FIG. 10–12. The thorium (4n) series.

12–4 The artificial radionuclides: alpha-emitters. It was mentioned in the last section that some of the isotopes of the transuranium elements decay by emission of an α-particle. In 1939, only 24 α-emitters were known and these were members of the naturally occurring radioactive series. At the present time, about 150 α-decaying nuclear species are known.(32) About 50 of these are among the transuranium elements; others form part of a new radioactive series, the $4n + 1$, or neptunium, series. There are several so-called *collateral series*, and additional α-emitters are found among the neutron-deficient isotopes of bismuth ($Z = 83$), polonium ($Z = 84$), astatine ($Z = 85$), emanation ($Z = 86$), and francium ($Z = 87$).

The members of the neptunium series(33,34) and their properties are listed in Table 12–1, and the relationships between the members are

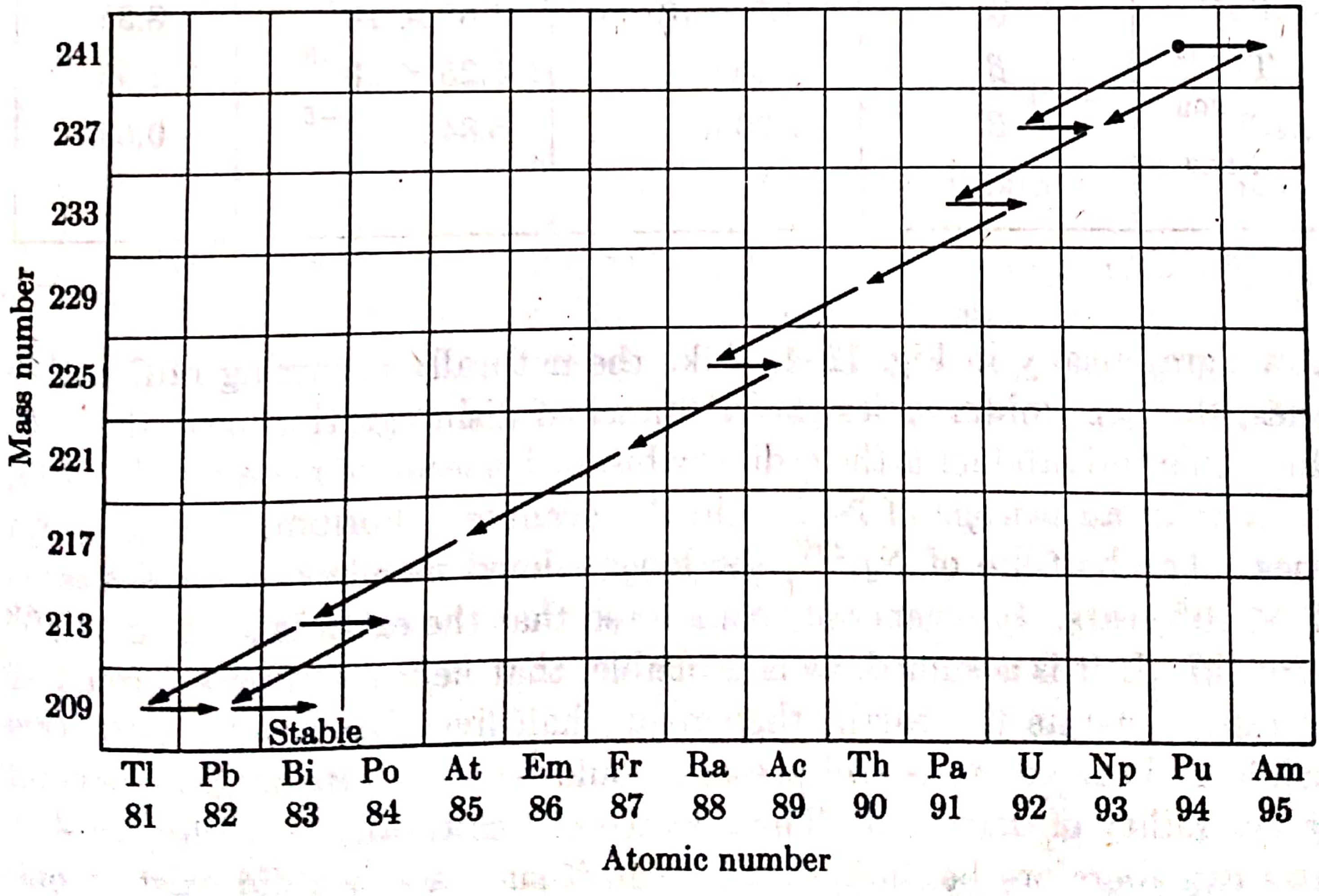


FIG. 12–4. The neptunium ($4n + 1$) series.

Absorption & Range of particles:

1) Alpha-particle (α):

The alpha particle is a massive mass in comparision to beta particle. It just can travel short distance in air medium and can easily be stopped by a thin sheet of paper & skin. These particles are easily absorbed by materials. They can travel a few centimeters in air. They typically penetrate skin about 40 micrometers of depth, equivalent to a few cells deep. They have shortest penetrating power since they have greatest capacity of ionization in the medium. (100 times greater than β particle & 10,000 times greater than γ-rays). Penetrating power is inversely proportional to their ionizing power. When α-particle passes through a medium, there is possibility to capture two electrons and becomes neutral helium atom. The distance the α-particle travels in the gas is called its range.

The distance through which an α-particle travels in a substance before coming to rest is called the 'range' of the particle in that substance.

The range of the α-particle depends on the following factors.

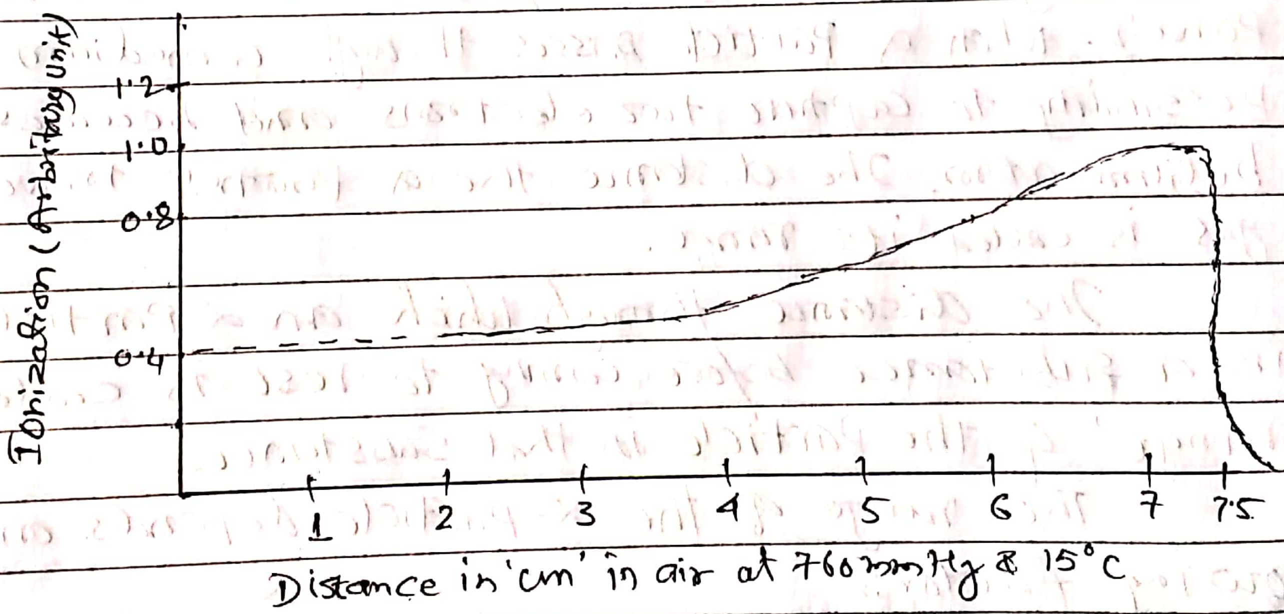
i) The radioactive substance giving out the α-particle
ii) The nature of the absorbing medium.
iii) The initial velocity of emission, the range being proportional to v^3 (Geiger's law).

The range of an α-particle can be measured

1) with the help of Wilson cloud chamber
2) by counting the number of ions produced along the path
3) by scintillations produced on a fluorescent screen.

Alpha particles are associated with energy from 4 Mev to 9 Mev.

It is experimentally established that the range is inversely proportional to the density of the gas ie. the range is inversely proportional to the pressure of the gas. Specific ionization is the number of ions produced by α-particles per unit length of the path. Bragg's research found that the curve relating to specific ionization with the distance from source was as shown in fig. It is seen that the graph a Bragg's hump near the end of the range and also a slight 'tail' or 'straggling effect'.



In above fig, the track length of the particles in the beam are not all the same, but vary around an average value. This effect is called straggling.

31.14 Theory of Alpha Decay

Only heavy nuclei with $A > 200$ undergo α-decay. The α-particles emitted from nuclei have a discrete energy spectrum and consist of several groups. Usually the most intensive is the group with α-particles of highest energy. The existence of several groups of α-particles is called the *fine structure of the α-spectrum.*

Before emission, the α-particle can be considered to be inside the nucleus. Coulomb's law is applicable when the α-particle is outside the nucleus. When the α-particle is inside the nucleus or very close to it, Coulomb's law does not hold good. For an α-particle of charge $2e$ and a nucleus of charge (Z–2) e separated by a distance r, the potential energy is given by

$$V_r = \frac{2e(Z-2)e}{4\pi\varepsilon_0 r} = \frac{2(Z-2)e^2}{4\pi\varepsilon_0 r}.$$

Here $(Z - 2)$ is the atomic number of the daughter nucleus. An attractive potential, called the potential well, represents the position (Fig. 31.12). When $r < r_0$, the α-particle is within the potential well and it is bound by the nucleus. Here r_0 is equal to the sum of the radii of the nucleus and the α-particle. According to wave mechanics, an α-particle can have different energy levels E_0, E_1, E_2, etc., within the potential well. Only an α-particle at a level E_3 or above can come out of the potential well. In the case of radioactive elements, the height of the potential barrier is about 9 MeV. But uranium emits α-particles of energy 4 MeV. Classical mechanics cannot explain how a particle having an energy of 4 MeV can come out of a well having a potential barrier of 9 MeV.

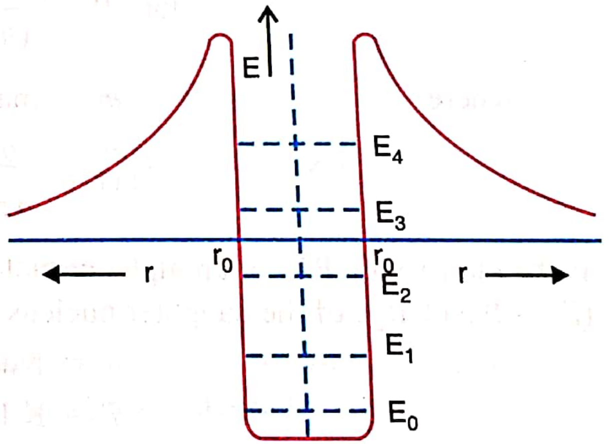
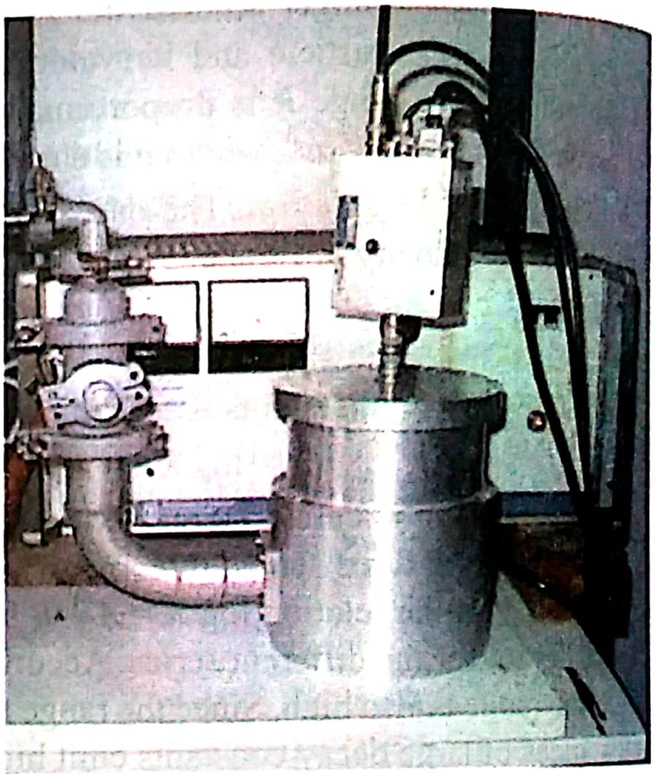


Fig. 31.12

The escape of an α-particle from a radioactive nucleus can be explained on the basis of wave mechanics and Schrodinger equation. According to it, it is possible for an α-particle to *leak* through the potential barrier, even though its K.E. is less than the potential energy of the height of potential barrier. This probability of leaking of an α-particle through the barrier is called *tunnel effect*. The α-particle within the nucleus must present itself again and again at the barrier surface until conditions are ripe for penetration or leakage.

31.13 Alpha Particle Spectra

It was thought earlier that the nucleus of a radioactive substance emitted alpha particles of some fixed energy. By measuring the energies of α-particles accurately using a magnetic spectrograph, it was shown that several radioactive nuclei emitted α-particles of different energies falling within a narrow range. As an example, consider a *ThC* nucleus which decays into a *ThC"* ($_{81}Tl^{208}$) nucleus after emitting an α-particle. *ThC* emits five groups of α-particles with energies 5.709, 5.728, 5.873, 6.161 and 6.201 MeV. The emission of groups of α-particles with different energies suggests that nuclei may exist in a number of discrete, excited energy states above the ground state. The parent nucleus decays from its ground state to anyone of the several excited states or to the ground state of the daughter nucleus. The daughter nucleus *Th C"* formed as a result of disintegration of *ThC* exists in several excited energy states *a*, *b*, *c*, *d*, *e*. *e* is the ground state and *a* is the highest excited state of the daughter nucleus. Each excited state has got some energy which it gives up in the form of γ radiation during its transition to the ground state. When a parent nucleus goes from its ground state to the ground state of the daughter nucleus, it emits during such a transition, an α-particle of maximum energy (α_1 in the energy diagram, Fig. 31.11). Any transition from the ground state of the parent nucleus to any one of the excited states of the daughter nucleus will give rise to α-particle with smaller energy (α_2, α_3, α_4 and α_5 in Fig. 31.11). The transition from any excited state to the ground state involves the emission of γ-rays. The energy of the γ-ray photons have beeen measured and found to be equal to the energy difference between one of the excited energy states and the ground state. Thus, the α-spectrum provides a conclusive evidence of the existence of *discrete energy levels in a nucleus.*



Setup for taking Alpha Particle Spectra.

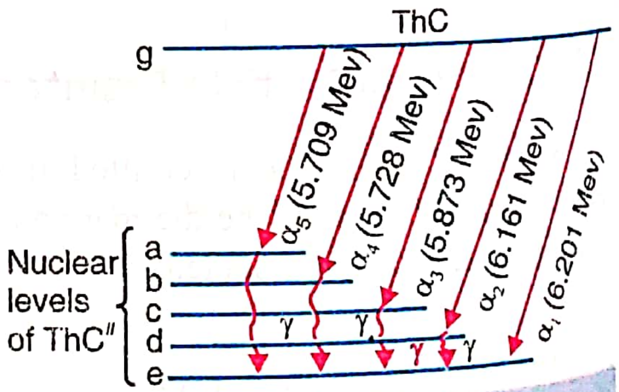


Fig. 31.11

Decay of parent nucleus ThC into various excited levels of the daughter nucleus ThC"

31.19 Beta Ray Spectra

The energies of β-particles from radio-active elements are determined by measuring the radii of curvature of their paths in a magnetic field of known flux density B. The circular path traversed by the β-particles of velocity v is governed by the relation

$$Bev = \frac{mv^2}{r}.$$

$$\therefore \quad v = Br\,(e/m).$$

From the geometry of the arrangement, the radius of the circular path r can be found.

The value of e/m can be assumed. Hence, the velocity v can be calculated.

For particles moving with very high velocities, the kinetic energy of the particle,

$$E_k = mc^2 - m_0c^2 = m_0c^2\left[\frac{1}{(1-v^2/c^2)^{1/2}} - 1\right]$$

31.22 The Neutrino Theory of Beta Decay

In 1934, Fermi developed a theory to explain the continuous β-ray spectrum. This theory is called neutrino theory of β-decay. According to this theory, a β-particle and a neutrino are created in the nucleus and both are emitted simultaneously. The *total energy* of these two particles is a constant which is equal to the end-point energy observed in the β-ray spectrum. This maximum energy is shared by the β-particle, the neutrino and also by the recoiling nucleus. The electron will carry the maximum energy when the energy of the neutrino is zero. In all other cases electron will carry an energy less than the maximum. The sum of the energies carried by the electron and the neutrino will always be the same. This energy may be shared by the two particles in any proportion. Hence it explains the continuous β-ray spectrum.

When the nucleon shifts from the neutron quantum state to the proton quantum state, electron and antineutrino are emitted. This process is represented by

$$n \rightarrow p + e^- + \bar{\nu}\,.$$

In ordinary beta decay it is an antineutrino that is emitted.

Positron emission corresponds to the conversion of a nuclear proton into a neutron, a positron, and a neutrino.

$$p \rightarrow n + e^+ + \nu.$$

Positron emission leads to a daughter nucleus of lower atomic number Z while leaving the mass number A unchanged. Thus negative and positive beta decays may be represented as

$$Z^{X^A} \rightarrow Z+1^{X^A} + e^- + \bar{\nu}$$

$$Z^{X^A} \rightarrow Z-1^{X^A} + e^+ + \nu$$

The electron, neutrino and product nucleus share among them the energy, angular momentum and linear momentum available from the nuclear transitions. Thus the neutrino theory of β-decay successfully explains the continuous energy spectrum of β-rays.

31.36 Biological Effects of Nuclear Radiations

The fact that nuclear radiation such as γ-rays, α-particles, β-particles, and neutrons can cause damage to the human body is well known.

The harmful effects of nuclear radiations appear to be due to the ionization or excitation of atoms in living cells by the Compton effect, bremsstrahlung, photoelectric effect, and so on. Some of the cell constituents are altered or destroyed by ionisation and some of the products formed may act as poisons. Examples of damages are the breaking up of chromosomes, swelling of the nucleus of a cell or of the entire cell; changes in the permeability of cell membrane, and destruction of cells.

The biological effects can be divided into three groups :

(*i*) Short-term recoverable effects;

(*ii*) Long-term irrecoverable effects; and

(*iii*) The genetic effect.

Groups (*i*) and (*ii*) are limited to the individuals who have actually received the radiation while the effects in group (*iii*) only appear in later generations.

When excessive doses are absorbed, the first noticeable disorder is a drop in the white blood cell count, which becomes evident in the first few hours after exposure. This is followed by a sickness pattern of diarrhoea,vomiting and fever which is now called radiation sickness. Recovery is possible from small acute doses. Large doses are lethal within a few weeks. Smaller doses produce short-term effects, such as skin disorders and loss of hair, which are generally recoverable. More serious is the damage done to the bone marrow and to other cells which may not have the ability to repair the damage. This leads to leukaemia and to the production of cancerous cells and finally malignant tumours.

The effects of the third type of damage only appear in the future generations of those irradiated. Experiments on animals show that the genetic effects to be expected from breeding from radiation-dosed human individuals include an increase in mental deficiencies, an increase in the number of monsters born and a general deterioration of the species in quality and population number.

The S.I. unit of radiation dosage is the *gray*. 1 *Gy* corresponds to 1 *J* of energy absorbed per kilogram of target material. The gray is a large unit and the *rad*, equal to 0.01 *Gy*, is more widely used. *One rad is defined as the absorbed dose of any nuclear radiation that results in the absorption of 0.01 J of energy per kilogram of absorbing material.* Since the absorbed dose required to produce a certain effect may be different for various types of radiation, this difference is expressed by the relative biological effectiveness (*RBE*) of the particular nuclear radiation. Finally, a dose unit for biological effect is the *rem* which is the dose in rads multiplied by the *RBE*. The *RBE* of 250 *keV* X-rays is taken as 1, and a dosage of 1 rad of such X-rays to a person is considered 1 *rem* (rad equivalent man). The *RBE*s of other X-rays and of gamma and beta rays are close to 1. But the *RBE* of fast neutrons is about 10 and that of 1-MeV alpha particles is about 25; 1 rad absorbed from such alpha particles therefore means a dosage of about 25 rem.

31.33 Units of Radioactivity

Generally we express the activity of a radioactive substance in terms of *curie* and its submultiples *millicurie* (mCi) and micro curie (μCi). The curie is defined as the quantity of a radioactive substance which gives 3.70×10^{10} disintegrations / second. This is the number of disintegrations per second per gram of radium. Another *unit rutherford* (*Rd*) is also used. It is defined as the quantity of a radioactive substance which gives 10^6 disintegrations / second. It is to be noted that for a substance with very short half-life period, very little of the substance is required for 1 curie of activity. On the other hand, for the substance with very long half-life peroid, a very large quantity of the substance is required for 1 curie of activity.